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META-ANALYSIS

Transcatheter vs Surgical Aortic Valve Replacement in Lower-Risk Patients

An Updated Meta-Analysis of Randomized Controlled Trials

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ABSTRACT

BACKGROUND Longer-term outcomes are especially important for lower-risk patients with severe aortic stenosis undergoing transcatheter aortic valve replacement (TAVR) or surgical aortic valve replacement (SAVR). Additional randomized data comparing TAVR and SAVR have recently become available.

OBJECTIVES The purpose of this study was to perform an updated systematic review with conventional pairwise meta-analyses and pooled survival analyses using reconstructed time-to-event individual participant data (IPD) including the totality of randomized evidence comparing longer-term clinical outcomes after TAVR and SAVR in lower-risk patients.

METHODS The prespecified primary endpoint was all-cause death. Key secondary endpoints included stroke and the composite of death or disabling stroke. Cox proportional hazards frailty regression and restricted mean survival time models were fitted using reconstructed time-to-event IPD. In sensitivity analyses, proportional odds models were fitted with frailty terms. Conventional pairwise meta-analyses were performed under random and fixed effects assumptions.

RESULTS Six trials enrolling 5,341 lower-risk patients were included with 2,717 randomized to TAVR and 2,624 randomized to SAVR (weighted mean follow-up of 35.7 months). At 5 years in the pooled survival analyses of reconstructed time-to-event IPD, TAVR was associated with a 20% reduction in the hazard of all-cause death (HR: 0.80; 95% CI: 0.66-0.97; $P = 0.02$) and a 19% reduction in the hazard of all-cause death or disabling stroke (HR: 0.81; 95% CI: 0.68-0.96; $P = 0.01$) compared with SAVR. There was no difference in stroke (HR: 0.97; 95% CI: 0.74-1.26; $P = 0.80$).

CONCLUSIONS In lower-risk patients, TAVR was associated with a reduced hazard of death and death or disabling stroke compared with SAVR, while rates of stroke were equivalent. Most patients have not yet undergone 5-year follow-up, and so these findings may change as further longer-term data become available. The present data are informative for lower-risk patients and treating clinicians, but further randomized trials and longer-term follow-up are required, particularly in younger patients. (JACC. 2025;85:926–940) © 2025 by the American College of Cardiology Foundation.

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Longer-term outcomes after transcatheter aortic valve replacement (TAVR) and surgical aortic valve replacement (SAVR) are of utmost importance in lower-risk patients. Event rates are naturally diminished in lower-risk patients; therefore, meta-analysis can be useful to pool results for low frequency but clinically meaningful endpoints, such as mortality, across trials that are individually underpowered for these. Previous meta-analyses focusing on lower-risk patients have combined randomized and nonrandomized studies[1] or included only 3 trials with 2,611 randomized participants.[2] Significant additional randomized data for lower-risk patients are now available, with longer-term follow-up of 3 prior trials[3-5] as well as publication of 2 new investigator-initiated trials.[6,7] These new data represent an approximate doubling of the randomized evidence base. We therefore performed an updated systematic review and meta-analysis comparing TAVR with SAVR incorporating the totality of the newly expanded randomized evidence base, focusing specifically on lower-risk patients and performing comprehensive survival analyses utilizing reconstructed time-to-event individual participant data (IPD).

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METHODS

The authors declare that all analyses included in this study are reported within the paper and appendix. Additional data and statistical code not presented are available from the corresponding author upon reasonable request. This study was registered with PROSPERO (CRD42023484976) and performed and reported in accordance with preferred reporting items for systematic reviews and meta-analyses (PRISMA) guidance[8] (Supplemental Table 1). Institutional Review Board approval or patient consent was not required because this was a study-level meta-analysis.

SEARCH STRATEGY AND DATA EXTRACTION. MEDLINE, Embase, and Cochrane databases from inception through June 2024 were systematically searched for randomized controlled trials (RCTs) reporting clinical endpoints that compared TAVR and SAVR. Search strings included ("severe aortic stenosis" OR "severe symptomatic aortic stenosis") AND ("transcatheter aortic valve implantation" OR "transcatheter aortic valve replacement") AND ("surgical aortic valve replacement" OR "aortic valve replacement") as used in a previously published systematic review,[9] with representative search strings shown in Supplemental Table 2. Bibliographies of previous meta-analyses, reviews, and conference abstracts from TCT, EuroPCR, American College of Cardiology, European Society of Cardiology, and American Heart Association meetings were manually searched. Searches were performed by 2 independent investigators (R.K.R. and Y.A.), and disputes were resolved by consensus. Representative search strings are presented in Supplemental Table 2. Data were extracted independently and in duplicate (R.K.R. and Y.A.). Additional data required for analyses that were not available in published trial reports were obtained from Principal Investigators (PIs) on request.

ABBREVIATIONS AND ACRONYMS

IPD = individual participant data

RCT = randomized controlled trial

RMST = restricted mean survival time

SAVR = surgical aortic valve replacement

TAVR = transcatheter aortic valve replacement

TREATMENT ARMS, INCLUSION AND EXCLUSION CRITERIA. Studies were eligible for inclusion if they randomized a population of lower-risk patients presenting with severe symptomatic aortic stenosis who were eligible for either TAVR or SAVR. The determination of lower risk was made by the study investigators on the basis of surgical risk, defined by the Society of Thoracic Surgeons Predicted Risk of Mortality. Observational studies were not eligible for inclusion. There were no language restrictions. No trials were excluded based on sample size or follow-up duration.

OUTCOMES. The prespecified primary endpoint was all-cause death; this was chosen because it is the most bias-resistant and clinically meaningful outcome. The main secondary endpoints were stroke and the composite of death or disabling stroke. If the composite of death or disabling stroke was not available, death or stroke could be used. Similarly, if only disabling stroke were available (as opposed to all stroke), this could be used for the stroke endpoint. Sensitivity analyses were performed including data with strict definitions of death or disabling stroke and all stroke.

Other endpoints were cardiovascular (or cardiac) death, new-onset atrial fibrillation, myocardial

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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infarction, new permanent pacemaker implantation, major or life-threatening bleeding, greater than mild and greater than moderate paravalvular regurgitation, rehospitalization, aortic valve reintervention, and acute kidney injury.

RISK OF BIAS ASSESSMENT. Risk of bias was assessed at the study level using Cochrane's Risk of Bias 2 Tool[10] for the following: 1) bias arising from the randomization process; 2) bias caused by deviations from the intended interventions; 3) bias caused by missing outcome data; 4) bias in measurement of the outcome; 5) bias in selection of the reported result; and 6) overall risk of bias. Sources of bias within domains were judged based on characteristics outlined in Cochrane's Handbook for Systematic Reviews of Interventions.

CERTAINTY OF EVIDENCE. The Grading of Recommendations, Assessment, Development, and Evaluations (GRADE) approach[11] was used to rate the quality of the evidence base underpinning the comparison of TAVR with SAVR.

STATISTICAL ANALYSIS. Conventional pairwise meta-analysis. For the conventional pairwise meta-analyses, endpoints were assessed as HRs over the entire follow-up duration by extracting study-level HRs and their corresponding 95% CIs. The natural logarithm of the HRs and their associated SEs were synthesized by employing random effects models fitted using the restricted maximum likelihood estimator[12] with the profile likelihood method used to derive CIs. The SE was calculated by dividing the difference between the natural logarithms of the upper and lower 95% CI by 2 times the appropriate normal score (1.96). Where the lower 95% CI approached zero, the SE was calculated using only the difference between the natural logarithm of the upper 95% CI and the natural logarithm of the point estimate. If HRs were not available, PIs were contacted to obtain this data. For the key endpoints (all-cause death, death or disabling stroke, and stroke), data from all trials were available in the form of HRs and 95% CIs. For the secondary endpoints (cardiovascular or cardiac death, new-onset atrial fibrillation, myocardial infarction, new permanent pacemaker implantation, major or life-threatening bleeding, greater than mild and greater than moderate paravalvular regurgitation, rehospitalization, aortic valve reintervention, and acute kidney injury) HRs were extracted for meta-analysis whenever they were reported. Event counts reported in 2×2 tables were extracted separately to calculate pooled relative risks (RRs) with corresponding 95% CIs at the longest reported follow-up, because HRs were not available for all secondary endpoints. HRs and RRs were not combined in a single meta-analysis for any endpoint.

To investigate the influence of each trial on the overall effect estimate, leave-one-out sensitivity analyses were performed, excluding each trial in turn. Higgins' and Thompson's I^2 statistic was calculated as an indirect measure of heterogeneity to assess the proportion of variance not explained by chance alone.[13] To directly quantify the presence of inter-study heterogeneity, Cochrane's Q test was performed, and Q statistics were calculated along with their corresponding *P* values.

Survival analyses using reconstructed time-to-event individual participant data. Prespecified pooled survival analyses were performed using reconstructed time-to-event IPD for the key outcomes of all-cause death, death or disabling stroke, and stroke. None of the secondary outcomes were studied in these analyses because of a paucity of data. Time-to-event IPD was reconstructed in a stepwise fashion. First, coordinate data was obtained via digitization of Kaplan-Meier (KM) plots. Summary-level data were extracted from individual trial reports (number of patients in each arm, number of events in each arm, and number of patients at risk at various time intervals). These were used in combination with the coordinate data from KM plots to produce time-to-event IPD, which detailed the survival time for each individual trial participant.[14,15] KM plots and Cox proportional hazards regression models were fitted using the time-to-event IPD. The log-rank test was performed to compare the survival distributions. To calculate HRs and their corresponding 95% CIs from the time-to-event IPD, unadjusted Cox models were constructed first. In recognition of the fact that differential baseline hazards exist among distinct randomized populations, a prespecified γ frailty term was included within the regression model to account for heterogeneity between trials, modeling trials as random effects using random intercepts.[16] The significance of the variance parameter was assessed with the likelihood ratio test. To test the proportional hazards assumption, scaled Schoenfeld residuals were graphed against the transformed time, and the Grambsch-Therneau test was performed.[17] For the primary analyses, follow-up was truncated at 5 years, because this was the longest duration at which at least 2 trials contributed time-to-event data to the analyses.[4,5] Construction of regression models that

allowed for time-varying HRs were planned, with follow-up truncated at yearly intervals to assess the temporal relationship between the comparison of TAVR and SAVR on the endpoints studied. Sensitivity analyses were conducted using these data with a proportional odds model fitted with a frailty term for study-level heterogeneity, modeled as a random intercept. These analyses are reported as OR with 95% CI.

Finally, to assess the total lifetime lost, restricted mean survival time (RMST) in patients randomized to TAVR or SAVR was calculated. RMST was calculated using the time-to-event IPD by performing 2-sample comparisons with RMST as a summary measure of the survival time distribution.[18] The difference in RMST is reported as the most clinically interpretable RMST metric. Differences in RMST are expressed in months with corresponding 95% CI.

Assessment of small-study effects, graphically using funnel plots and statistically using linear regression, was planned if at least 10 studies were eligible for synthesis.[19] Analyses were performed within the as-randomized samples, in accordance with the intention-to-treat principle, where data were available. Significance testing was performed at the 2-tailed 5% significance level. The statistical programming environment R version 4.3.1 was used for all analyses.

RESULTS

Six trials were eligible for this meta-analysis,[3-7,20] after initial searches identified 1,268 studies. The search results are presented in Supplemental Figure 1. A total of 5,341 lower-risk patients were included, with 2,717 randomized to TAVR and 2,624 randomized to SAVR. The weighted mean follow-up duration across all included trials was 35.7 months. The characteristics of the included trials are summarized in **Table 1**. Demographics of the trial populations are summarized in Supplemental Table 3. Cochrane's Risk of Bias 2 assessment of each trial is reported in Supplemental Table 4. Two trials were judged to be at low risk of bias.[5,7] Four trials had some concerns for bias caused by deviations from intended interventions.[3,4,6,20]

The certainty of the evidence was high for the primary endpoint (all-cause death) and key secondary endpoints (death or disabling stroke and stroke). The Summary of Findings GRADE table is presented in Supplemental Table 5. Definitions of outcomes used in each individual trial are reported in Supplemental Table 6. Quality assessment of the reconstructed time-to-event IPD is provided in Supplemental Tables 7 and 8. The fidelity of the data reconstruction was very high. In terms of event counts, for all-cause death there was 99.6% accuracy, for death or disabling stroke there was 97.8% accuracy and for stroke there was 95.3% accuracy. Similarly, when we recreated the original trial Cox proportional hazards regression analyses, the HRs were nearly identical, and no major differences were noted during a visual comparison between the originally published KM plots and the reconstructed KM plots for all endpoints in each individual trial. Additional data included in the analysis not available from published trial reports included KM plots for stroke and death or disabling stroke and their corresponding HRs, from the NOTION (Nordic Aortic Valve Intervention) trial and the KM plot for death or disabling stroke from the UK-TAVI (UK Transcatheter Aortic Valve Implantation) trial. These were provided by the PIs upon request.

ALL-CAUSE DEATH. In the conventional pairwise random-effects meta-analyses across the 6 lower-risk trials reporting outcome data for all-cause death, the HR with TAVR compared with SAVR was 0.82 (95% CI: 0.60-1.13; $P = 0.23$). There was substantial statistical heterogeneity ($I^2 = 64.3\%$). In fixed effects models, the results remained consistent (HR: 0.86; 95% CI: 0.73-1.01; $P = 0.07$; $I^2 = 60.8\%$) (**Figure 1**). Forest plots for the conventional pairwise meta-analyses on all-cause death are presented in **Figure 1**. Forest plots for the leave-one-out sensitivity analyses are presented in Supplemental Figure 2.

In the Cox frailty model fitted with time-to-event IPD, TAVR was associated with a 20% reduction in the hazard of all-cause death at 5 years compared with SAVR (HR: 0.80; 95% CI: 0.66-0.97; $P = 0.02$). The proportional hazards assumption was met (Schoenfeld $P = 0.49$). The variance of the random effect is reported in Supplemental Table 9. HRs derived at 1, 2, 3, and 4 years are presented in Supplemental Table 10 with consistent results at every time point. Results for the unadjusted Cox model are presented in Supplemental Table 11. Similarly, in the sensitivity analysis that fitted a proportional odds model, the odds of death were lower with TAVR compared with SAVR (OR: 0.79; 95% CI: 0.64-0.96; $P = 0.02$). The RMST free of all-cause death was 1 month greater with TAVR than with SAVR (56.3 months vs 55.3 months; $P = 0.02$). The pooled KM plot for all-cause death at 5 years is presented in **Figure 2**.

ALL-CAUSE DEATH OR DISABLING STROKE. In the conventional pairwise random-effects meta-analyses across the 6 lower-risk trials reporting outcome data

TABLE 1 Characteristics of Included Trials

First Author	Study	Year	Region	N	Mean Age, y[a]	Female, %	Follow-Up[b]
Toff et al[20]	UK TAVI	2022	United Kingdom	913	81.0 ± 4.4	46.4	1
Forrest et al[3]	Evolut Low Risk	2023	North America, Australia, Europe, Japan	1,414	74	35.3	4
Jørgensen et al[5]	NOTION	2023	Denmark, Sweden	280	79.1 ± 4.8	46.8	10
Mack et al[4]	PARTNER 3	2023	United States, Australia, New Zealand, Japan	1,000	73.3 ± 5.8	30.7	5
Blankenberg et al[6]	DEDICATE-DZHK6	2024	Germany	1,414	74 ± 4	42.7	1
Jørgensen et al[7]	NOTION-2	2024	Denmark, Norway, Sweden, Finland and Iceland	376	71.1	37.3	1

[a]Mean age in years ± SD; value for TAVR group provided where values differ between transcatheter aortic valve replacement (TAVR) and surgical aortic valve replacement (SAVR) groups, and overall value not reported. [b]Follow-up in years. [c]Further details of definitions are provided in Supplemental Table 3.

6MWT = 6-min walk test; AF = atrial fibrillation; AR = aortic regurgitation; AS = aortic stenosis; AV = aortic valve; AVA = aortic valve area; CV = cardiovascular; DEDICATE-DZHK6 = Randomized, Multicenter, Event-Driven Trial of TAVI versus SAVR in Patients with Symptomatic Severe Aortic-Valve Stenosis; Evolut Low Risk trial = Evolut Surgical Replacement and Transcatheter Aortic Valve Implantation in Low Risk Patients; KCCQ = Kansas City Cardiomyopathy Questionnaire; LV = left ventricle; LVEF = left ventricular ejection fraction; MACE = major adverse cardiac event; MI = myocardial infarction; MLWHF = Minnesota Living With Heart Failure Questionnaire; NOTION = Nordic Aortic Valve Intervention; PARTNER 3 = Safety and Effectiveness of the SAPIEN 3 Transcatheter Heart Valve in Low Risk Patients With Aortic Stenosis; PG = pressure gradient; PPM = permanent pacemaker; SF-12 = Medical Outcomes Study 12-Item Short Form General Health Survey; STS-PROM = Society of Thoracic Surgeons Predicted Risk of Mortality; UK TAVI = UK Transcatheter Aortic Valve Implantation.

Continued on the next page

for all-cause death or disabling stroke, the HR for TAVR compared with SAVR was 0.84 (95% CI: 0.61-1.14; $P = 0.26$). There was substantial statistical heterogeneity ($I^2 = 65.4\%$). In fixed effects models, TAVR associated with a statistically significant 27% reduction (HR: 0.83; 95% CI: 0.71-0.98; $P = 0.03$; $I^2 = 63.0\%$) (**Figure 3**). Forest plots for the conventional pairwise meta-analyses on death or disabling stroke are presented in **Figure 3**. Forest plots for the leave-one-out sensitivity analyses are presented in Supplemental Figure 2.

In the Cox frailty model fitted with time-to-event IPD, TAVR was associated with a 19% reduction in the hazard of death or disabling stroke at 5 years compared with SAVR (HR: 0.81; 95% CI: 0.68-0.96; $P = 0.01$). The proportional hazards assumption was met (Schoenfeld $P = 0.32$). The variance of the random effect is reported in Supplemental Table 9. HRs derived at 1, 2, 3, and 4 years are presented in Supplemental Table 10 with consistent results at every time point. Results for the unadjusted Cox model are presented in Supplemental Table 11. Similarly, in the sensitivity analysis that fitted a proportional odds model, the odds of death or disabling stroke were lower with TAVR compared with SAVR (OR: 0.80; 95% CI: 0.66-0.96; $P = 0.01$).

TABLE 1 Continued

Entry Criteria	STS-PROM Score, %	TAVR Type	Primary Outcome[c]	Secondary Outcomes[c]
Severe, symptomatic AS Age ≥80 or ≥70 y with intermediate or high risk as determined by multidisciplinary team	2.6	Any CE marked valve	All-cause death	All-cause death at 2, 3, 4, and 5 y. Stroke, composite of all-cause death or stroke, composite of all-cause death or disabling stroke, conduction disturbance requiring permanent pacemaker, endocarditis, reintervention, vascular complications, major bleeding, renal replacement therapy, MLWHF, EQ-5D-5L, NYHA, 6MWT, TTE measures.
Severe, symptomatic AS AVA ≤1.0 cm^2 (index 0.6 cm^2/m^2) with peak velocity ≥4 m/s or mean PG ≥40 mm Hg. Asymptomatic if LVEF <50%. Asymptomatic: peak velocity ≥5 m/s or mean PG ≥60 mm Hg or LVEF <50% or abnormal exercise test. Low risk for SAVR (Heart Team agreed STS PROM score ≤3%).	1.9	Medtronic CoreValve Evolut R Evolut PRO Self-expanding, porcine pericardium	Composite of death or disabling stroke	All-cause death, disabling stroke, life-threatening bleed, major vascular complications, stage II or III acute kidney injury (composite), new PPM at 30d. Endocarditis, thrombosis, all stroke, life threatening bleeding, reintervention at 1yr. Mean AV gradient, AVA, NYHA, KCCQ.
Severe, symptomatic AS AVA ≤1 m^2 (index 0.6 cm^2/m^2) and either peak velocity >4 m/s or mean PG >40 mm Hg. Asymptomatic if LV posterior wall thickness ≥17 mm.	<4	Medtronic CoreValve Self-expanding, porcine pericardium	Composite of death, stroke, and MI	Primary outcome components, cardiovascular death, prosthesis reintervention, cardiogenic shock, endocarditis, pacemaker, atrial fibrillation/flutter, vascular, renal, bleeding complications, AVA, AV gradient, AR.
Severe, symptomatic AS AVA ≤1.0 cm^2 (index 0.6 cm^2/m^2) with peak velocity ≥4 m/s or mean PG ≥40 mm Hg. Asymptomatic if LVEF <50% or abnormal exercise test. Low risk for SAVR (heart team agreed STS PROM score <4%).	1.9	Edwards Sapien 3 Balloon expandable, bovine pericardium	Composite of death, stroke or rehospitalization related to the valve, the procedure, or heart failure	Stroke, composite of death or stroke, new-onset AF at 30 d, hospitalization duration, composite of death or low KCCQ at 30 d. NYHA, KCCQ, 6MWT.
Severe, symptomatic AS and at least age 65 years Low or intermediate surgical risk according to heart team	1.8	Any CE marked valve	Composite of death or fatal or nonfatal stroke	Components of the primary outcome along with acute kidney injury, arrhythmia and PPM implantation, bleeding, MI prosthetic-valve dysfunction, rehospitalization, and vascular complications.
Severe symptomatic AS or asymptomatic with abnormal exercise test showing symptoms on exercise clearly related to AS and ≤75 years of age Severe calcific tricuspid or bicuspid AS (aortic valve area <1.0 cm^2 or <0.6 cm^2/m^2 and mean transvalvular gradient >40 mm Hg or peak jet velocity >4.0 m/s) STS PROM score <4%	1.1	Any CE marked valve	Composite of death or stroke or rehospitalization related to the procedure, valve or heart failure	All-cause death, stroke (disabling and nondisabling), major or life-threatening bleeding, new-onset atrial fibrillation, need for PPM implantation, valve endocarditis, valve thrombosis, need for valve reintervention and valve performance as assessed by transthoracic echocardiography.

The RMST free of death or disabling stroke was 1.1 months greater with TAVR than with SAVR (55.1 months vs 54 months; $P = 0.02$). The pooled KM plot for death or disabling stroke at 5 years is presented in **Figure 4**.

ALL STROKE. In the conventional pairwise random-effects meta-analyses across all 6 lower-risk trials reporting outcome data for stroke, the HR for TAVR compared with SAVR was 0.93 (95% CI: 0.60-1.45; $P = 0.96$). There was substantial statistical heterogeneity ($I^2 = 60.4\%$). Forest plots for the random effects meta-analyses on stroke are presented in **Figure 5**. In fixed effects models, the results remained consistent (HR: 0.86; 95% CI: 0.66-1.12; $P = 0.26$; $I^2 = 59.1\%$) (**Figure 5**). Forest plots for the conventional pairwise meta-analyses on stroke are presented in **Figure 5**. Forest plots for the leave-one-out sensitivity analyses are presented in Supplemental Figure 2.

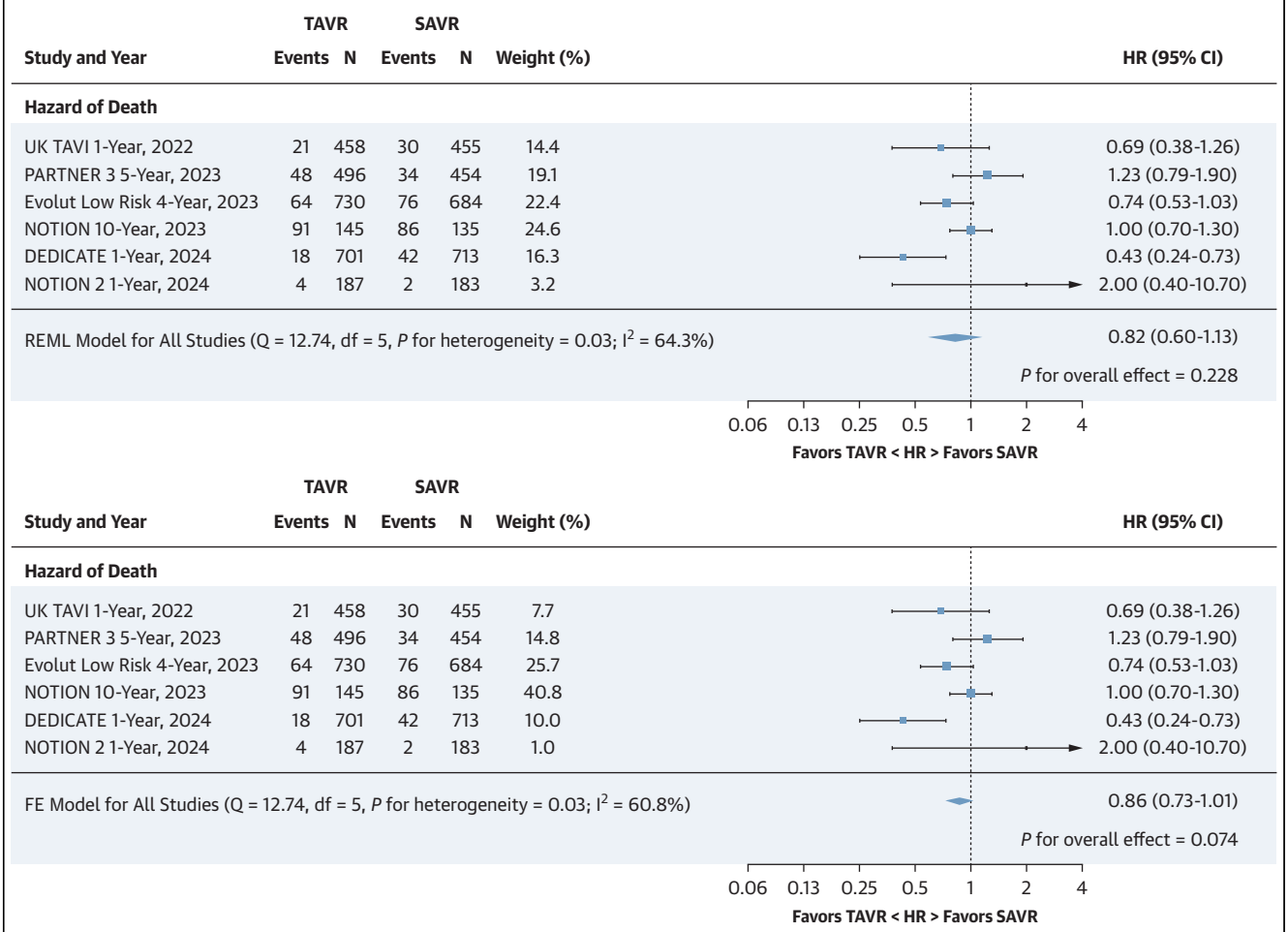
In the Cox frailty model fitted with time-to-event IPD, the equivalence of TAVR and SAVR was retained (HR: 0.97; 95% CI: 0.74-1.26; $P = 0.80$). The proportional hazards assumption was met (Schoenfeld $P = 0.13$). The variance of the random effect is reported in Supplemental Table 9. HRs derived at 1, 2, 3, and 4 years are presented in Supplemental Table 10 with consistent results at every time point. Results for the unadjusted Cox model are presented in Supplemental Table 11. Similarly, in the sensitivity analysis that fitted a proportional odds model, the odds of stroke were similar with TAVR compared with SAVR (OR: 0.97; 95% CI: 0.74-1.27; $P = 0.80$).

The RMST free of all-stroke was similar between TAVR and SAVR (57.7 months vs 57.6 months; $P = 0.80$). The pooled KM plot for stroke at 5 years is presented in **Figure 6**.

A summary of the HR meta-analyses reported in the previous text are presented in the **Central Illustration**.

SECONDARY OUTCOMES. There were no significant differences in hazard between TAVR and SAVR in

FIGURE 1 Meta-Analysis on All-Cause Death Comparing TAVR With SAVR



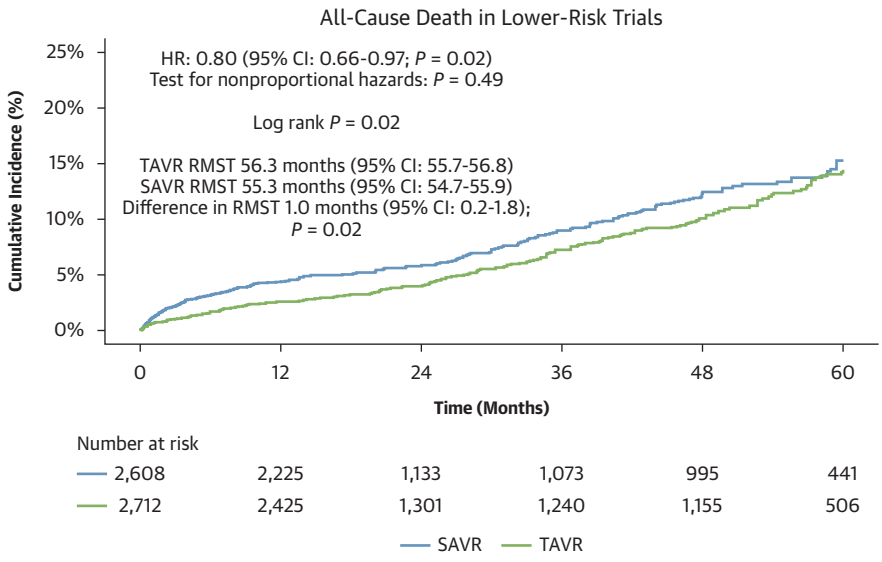
(Top) Random effects HR meta-analysis. (Bottom) Fixed effects HR meta-analysis. REML = restricted maximum likelihood; SAVR = surgical aortic valve replacement; TAVR = transcatheter aortic valve replacement; FE = fixed effects.

disabling stroke (HR: 0.71; 95% CI: 0.46-1.10; $P = 0.12$; $I^2 = 13.2\%$), cardiovascular death (HR: 0.82; 95% CI: 0.49-1.35; $P = 0.43$; $I^2 = 42.2\%$), myocardial infarction (HR: 0.62; 95% CI: 0.38-1.02; $P = 0.43$; $I^2 = 0\%$), rehospitalization (HR: 0.83; 95% CI: 0.66-1.03; $P = 0.09$; $I^2 = 0\%$), and aortic reintervention (HR: 1.12; 95% CI: 0.63-2.00; $P = 0.69$; $I^2 = 4.1\%$).

Compared with SAVR, TAVR was associated with lower hazards of major or life-threatening bleeding (HR: 0.35; 95% CI: 0.23-0.54; $P < 0.001$; $I^2 = 79.5\%$), atrial fibrillation (HR: 0.21; 95% CI: 0.09-0.50; $P < 0.001$; $I^2 = 93.1\%$), and acute kidney injury (HR: 0.50; 95% CI: 0.26-0.95; $P = 0.03$; $I^2 = 0\%$). Compared with SAVR, TAVR was associated with an increased hazard of permanent pacemaker implantation (HR: 1.75; 95% CI: 1.41-2.16; $P < 0.001$; $I^2 = 7.3\%$), vascular complications (HR: 5.16; 95% CI: 2.16-12.31; $P < 0.001$; $I^2 = 57.0\%$), greater than mild paravalvular regurgitation (RR: 3.90; 95% CI: 2.82-5.40; $P < 0.001$; $I^2 = 65.5\%$) and greater than moderate paravalvular regurgitation (RR: 3.77; 95% CI: 1.78-7.99; $P < 0.001$; $I^2 = 0\%$).

The equivalent HR fixed effects, RR random effects and RR fixed effects meta-analysis forest plots are presented in Supplemental Figures 3 and 4.

FIGURE 2 Kaplan-Meier Plot for All-Cause Death at 5 Years Comparing TAVR With SAVR



RMST = restricted mean survival time; other abbreviations as in **Figure 1**.

SENSITIVITY ANALYSES. The results of the leave-one-out sensitivity analyses for the pairwise HR meta-analyses are presented in Supplemental Figure 2.

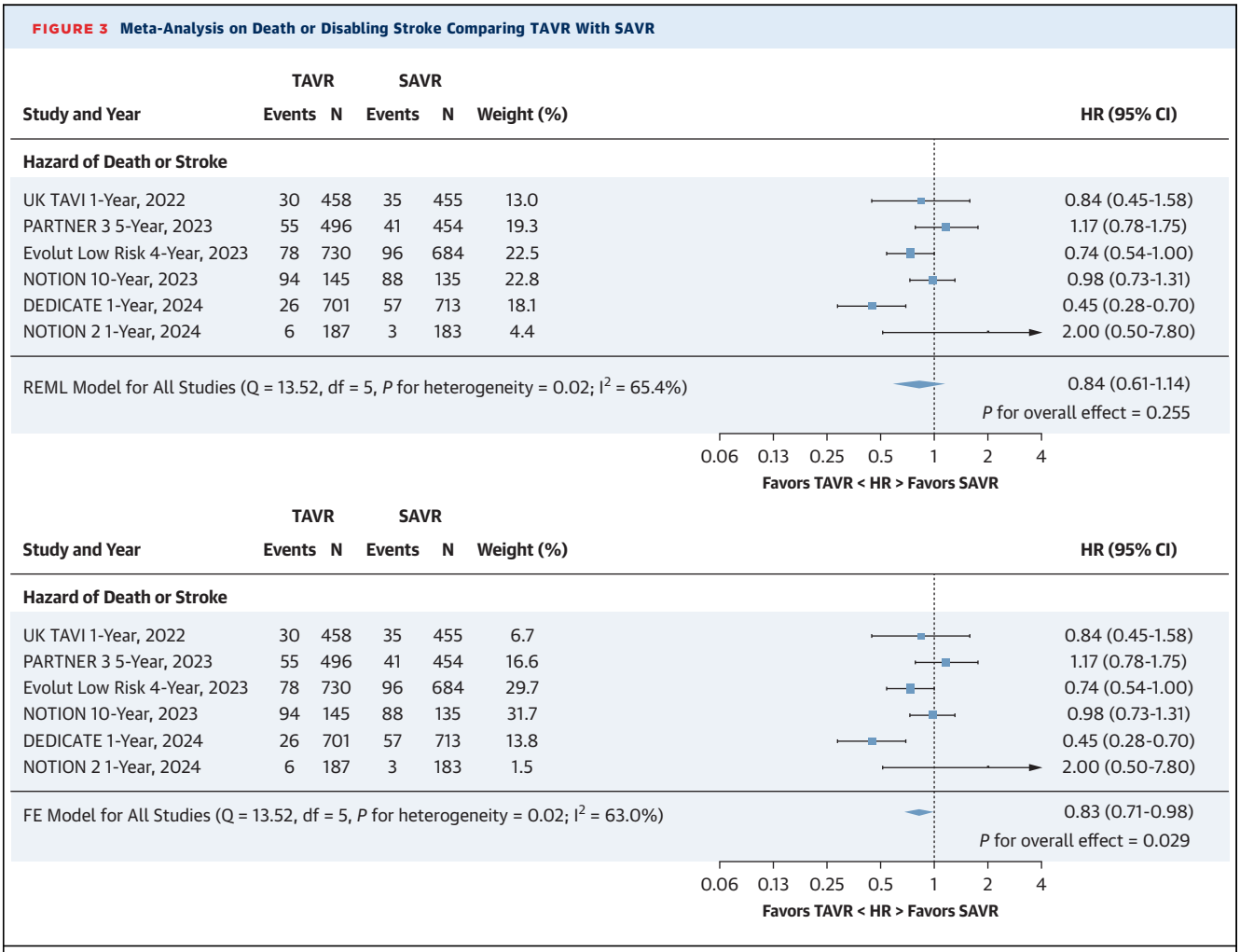
A nonprespecified subgroup analysis stratified by TAVR valve type was performed. There were no heterogenous treatment effects related to the TAVR valve employed, although these analyses were underpowered. The results of this analysis are presented in Supplemental Figure 5.

Three nonprespecified sensitivity analyses in the reconstructed time-to-event IPD were performed. The first excluded the NOTION trial[5] because of the significant presence of Trifecta valves, which were recently removed from the U.S. market following concerns about high rates of early structural valve deterioration. The second excluded the UK-TAVI trial,[20] because this trial could be considered to contain a mix of low and intermediate-risk participants. The results of these analyses were concordant with the primary analyses across the entire 6 trials. Third, it was suggested to exclude the studies from death or disabling stroke and stroke analyses where KM plots reporting slightly different endpoints were substituted. This resulted in the exclusion of the DEDICATE-DZHK6 (Randomized, Multicenter, Event-Driven Trial of TAVI versus SAVR in Patients with Symptomatic Severe Aortic-Valve Stenosis) trial[6] from the death or disabling stroke endpoint and exclusion of the Evolut Low Risk (Evolut Surgical Replacement and Transcatheter Aortic Valve Implantation in Low Risk Patients) trial[3] from the stroke endpoint. Results from Cox proportional hazards regression analyses and the accompanying KM plots excluding these trials are presented in Supplemental Figure 6, with similar results to the primary analyses.

DISCUSSION

The principal findings of this study are as follows: 1) using reconstructed time-to-event IPD in survival analysis models, TAVR is associated with a 20% reduction in all-cause death at 5-year follow-up (HR: 0.80; 95% CI: 0.66-0.97; $P = 0.02$); 2) using the same methodology, there was also a 19% reduction in the risk of death or disabling stroke at 5 years (HR: 0.81; 95% CI: 0.68-0.96; $P = 0.01$) and there were no differences in stroke rates; 3) these results were consistent when analyzed with various methodologies (Cox proportional hazards regression, sensitivity analyses with proportional odds models, RMST, and conventional pairwise meta-analyses with both random and fixed effects); and 4) consistent with prior work,[9] there was an increased risk of pacemaker implantation and paravalvular regurgitation after TAVR, but lower risks of acute kidney injury, bleeding, and new atrial fibrillation with no differences in cardiac death, myocardial infarction, aortic reintervention, or rehospitalization.

FIGURE 3 Meta-Analysis on Death or Disabling Stroke Comparing TAVR With SAVR

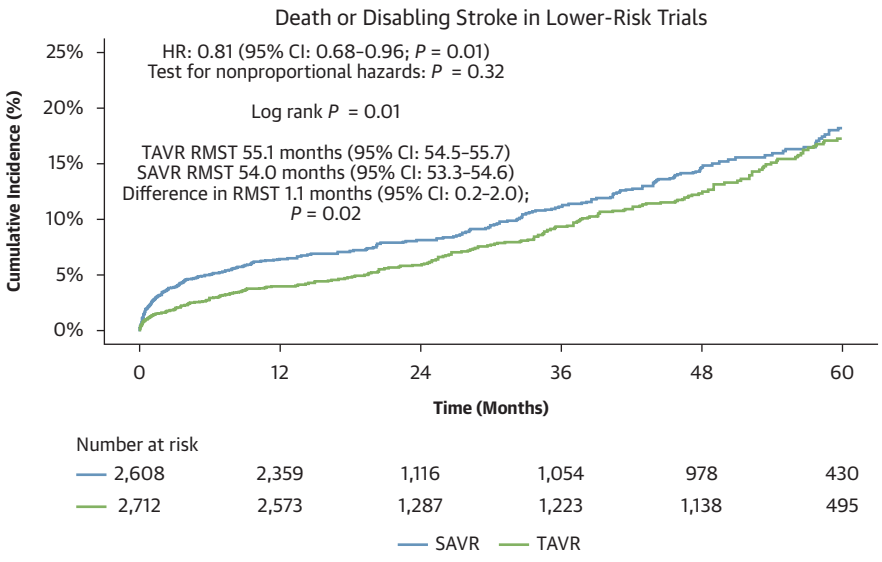


(Top) Random effects HR meta-analysis. (Bottom) Fixed effects (FE) HR meta-analysis. Abbreviations as in Figure 1.

Our analysis differs from previously published meta-analyses[1,2,9] in the following ways: 1) we have incorporated the totality of randomized data for lower-risk patients undergoing TAVR, with an overall 2-fold increase in the randomized evidence base compared with prior meta-analyses of RCTs;[2] 2) we incorporated this additional data from 2 new trials,[6,7] as well as 10-year follow-up from NOTION,[5] 5-year follow-up from PARTNER 3 (Placement of Aortic Transcatheter Valves 3),[4] and 4-year follow-up from Evolut Low Risk;[3] 3) using reconstructed participant time-to-event IPD analyses, we are able to demonstrate statistically significant reductions in death and the composite of death or disabling stroke with TAVR at 5-year follow-up in lower-risk patients, which have not been reported in prior meta-analyses of lower-risk RCTs;[2] 4) we have used various analytic methods to ensure consistency and confidence in our results; and 5) we focused only on randomized trials, in contrast to prior work that included observational studies and suggested a higher-risk of all-cause mortality with TAVR.[1] The certainty of evidence for our analysis assessed with the GRADE approach was high. An important caveat of the current study is that patients enrolled in the UK-TAVI, DEDICATE-DZHK6, and NOTION-2 trials did not contribute survival data beyond 1 year, and the majority of patients included in this analysis have not yet undergone 5-year follow-up. As further long-term data accumulate, these results may change. It is therefore critical that this analysis is updated with contemporaneous randomized data with longer-term follow-up as these become available.

With the conventional pairwise meta-analyses, the numerical reductions in death with TAVR did not reach conventional thresholds for statistical significance due to imprecision. However, in pooled survival analyses using reconstructed time-to-event IPD, TAVR was associated with a statistically significant

FIGURE 4 Kaplan-Meier Plot for Death or Disabling Stroke at 5 Years Comparing TAVR With SAVR



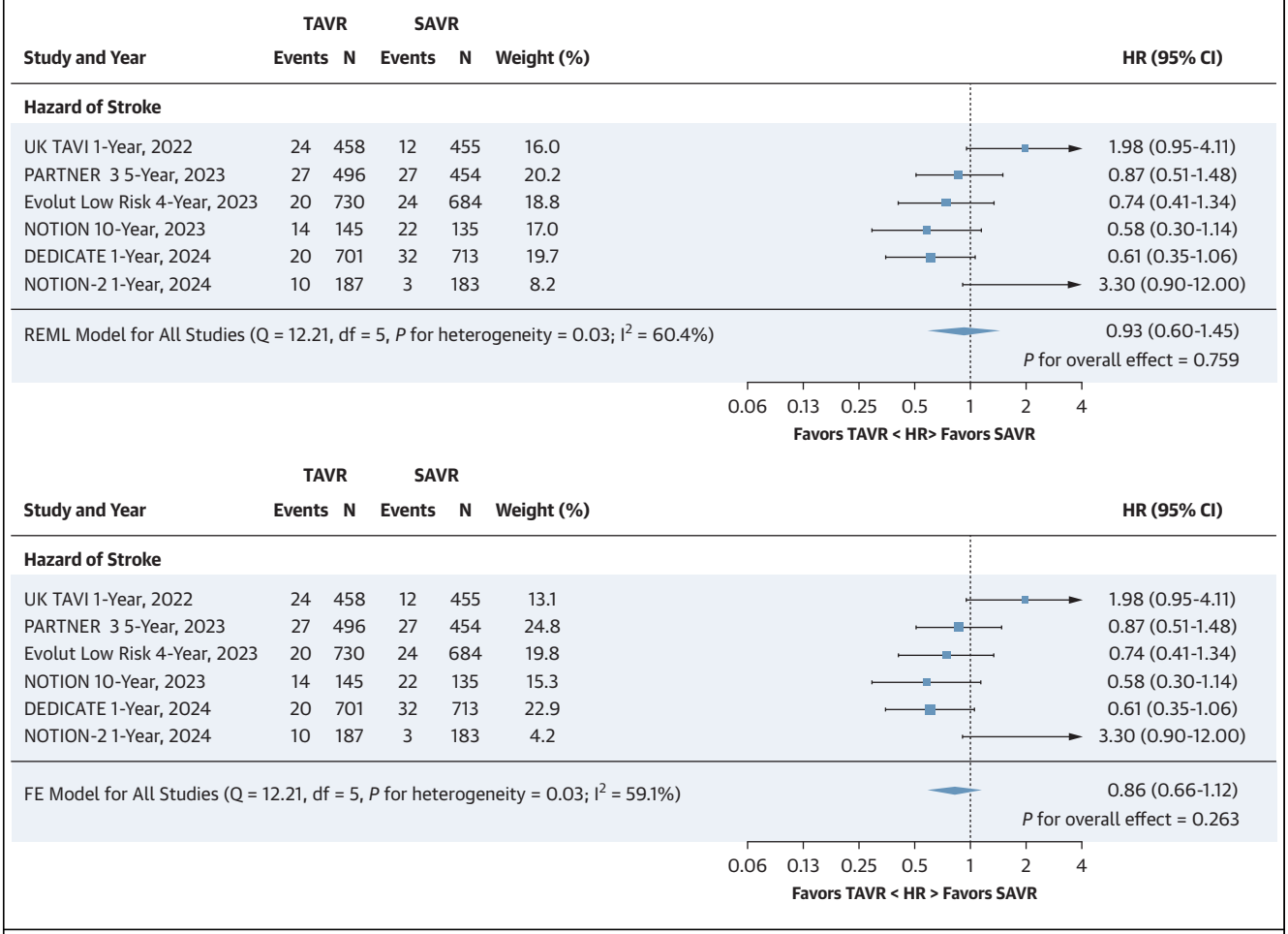
Abbreviations as in Figures 1 and 2.

20% reduction in death at 5 years compared with SAVR. These analyses have the advantage of leveraging more detailed time-to-event data from individual KM plots, enhancing the precision of the point estimates. Cox proportional hazards regression models were used, which are an appropriate method for summarizing time-to-event data. The use of frailty terms allowed us to account for heterogeneity in terms of differential baseline hazards in the distinct randomized populations. By reconstructing time-to-event IPD, we can therefore mitigate some of the limitations inherent in aggregated data approaches, such as loss of granularity and inability to fully account for time-dependent effects. Our results remained directionally consistent and statistically significant in sensitivity analyses where proportional odds models were fitted using reconstructed individual participant time-to-event data. Additionally, across 5 years, TAVR was associated with significantly longer RMST free of death and death or disabling stroke compared with SAVR, although the absolute increases were modest.

Our findings only apply to the patients enrolled in the contributing trials. Although these were lower-risk patients (compared with the foundational trials of TAVR), the majority of patients enrolled in these trials were in their 70s and 80s (median age in UK TAVI 81 years, in NOTION 79 years, in PARTNER 3 73 years, in Evolut Low-Risk 74 years, in DEDICATE-DZHK6 74 years, and in NOTION-2 71 years). Our results therefore do not apply to, and cannot be extrapolated to, patients in their 50s and 60s. A recent propensity-matched analysis from the National Inpatient Sample[21] showed that in patients aged 50 to 64 years with symptomatic severe aortic stenosis, TAVR was the mode of aortic valve replacement for 41.8% of patients in 2020 (increased from 14.2% in 2016). This is not an approach currently supported by the available randomized data (including this analysis), and before TAVR is adopted for very young patients, dedicated randomized trials would need to be conducted with long-term (~10 years) follow-up and consideration given to the lifetime management of these patients. Lifetime management of AS patients is of paramount importance, and our analysis cannot speak to issues of valve durability, aortic valve reintervention, or future coronary revascularization procedures for these patients. Clinicians must weigh all of those, and myriad other, factors when determining the optimal therapeutic strategy for lower-risk patients with severe AS.

STUDY LIMITATIONS. As a study-level meta-analysis, we did not have the requisite individual demographic data to perform analyses examining treatment-covariate interactions. Specifically, in this study, TAVR was associated with higher permanent pacemaker implantation and paravalvular leak, and prior

FIGURE 5 Meta-Analysis on Stroke Comparing TAVR With SAVR

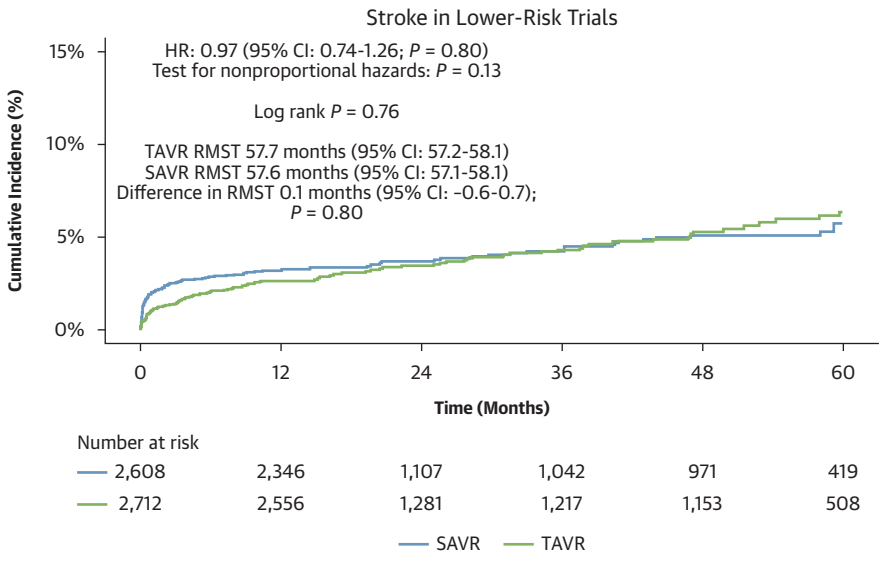


(Top) Random effects HR meta-analysis. (Bottom) Fixed effects HR meta-analysis. Abbreviations as in Figures 1 and 3.

work has demonstrated these outcomes could be associated with mortality.[22,23] Leveraging true individual patient data would be useful to study these issues with interaction testing, as study-level meta-regression can be unreliable due to ecologic fallacy and overfitting (particularly when k <10, as in our study). Similarly, both valve replacement strategies are affected by prosthesis-patient mismatch to varying degrees, which also correlates with mortality.[24,25] Prosthesis-patient mismatch was not uniformly defined or reported after either TAVR or SAVR in the constituent trials of this analysis, and therefore, we were unable to systematically study this. This is an important data point to capture, and future studies should aim to standardize definitions and reporting. The impact of pacemaker implantation, paravalvular leak, and prosthesis-patient mismatch on clinical endpoints could be explored within the framework of a true patient-level pooled analysis.

Common to most meta-analyses, there was lack of uniformity in follow-up duration, eligibility criteria, or endpoint definitions and subgroups across all included trials. Certain meta-analyses had evidence of increased statistical heterogeneity, although strict thresholds for interpreting statistical heterogeneity are not advised.[26] In the context of the significant heterogeneity in the conventional pairwise meta-analyses for the primary endpoint and key secondary endpoints, the results of the random effect models should be considered more reliable, although we also present fixed effect models for completeness and comparison. Detailed exploration of potential sources of heterogeneity could be achieved with true individual participant data and would be informative

FIGURE 6 Kaplan-Meier Plot for Stroke at 5 Years Comparing TAVR With SAVR



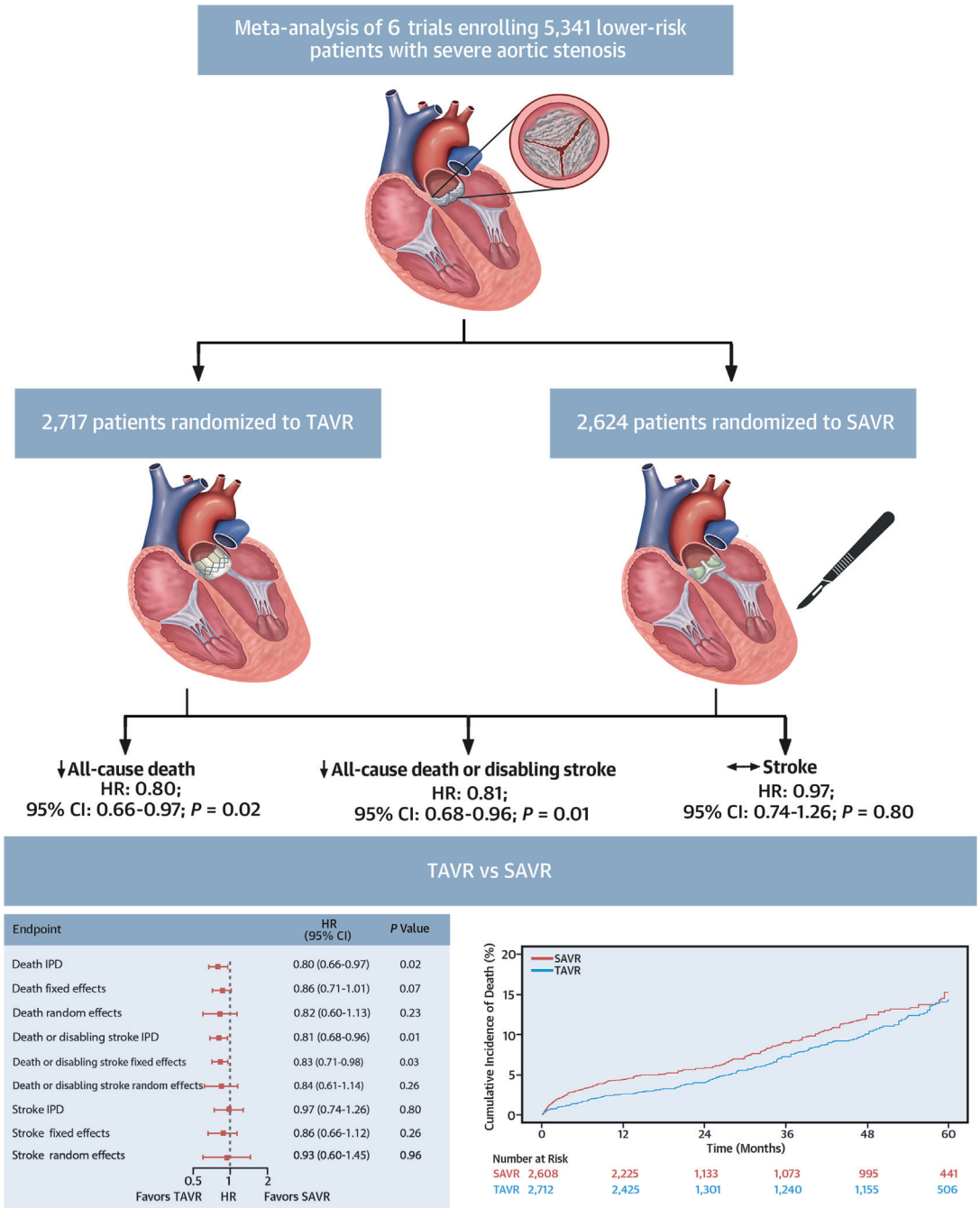
Abbreviations as in Figures 1 and 2.

for understanding the differences in estimated treatment effects. Our frailty term survival analyses on reconstructed time-to-event IPD were designed to account for between-trial heterogeneity.

Some investigators have criticized the RCTs comparing TAVR and SAVR for various issues, principal of which are loss to follow-up and concomitant additional procedures performed (mainly coronary revascularization in the form of CABG as part of the SAVR arm).[27] We did not consider any trials included in our analysis to be at high risk of bias. With regards to additional procedures, it is true that concomitant CABG is performed more commonly with SAVR than PCI is with TAVR; this practice is in accordance with guideline recommendations. The optimal management of concomitant severe CAD and severe AS can only be answered with adequately powered and designed RCTs. The recent TCW (Transcatheter Valves and Vessels) trial[28] randomized patients with severe AS and severe CAD to either FFR-guided PCI followed by TAVR or combined SAVR and CABG. The trial was stopped early because of excess events in the surgical arm (all-cause death and stroke occurred in 12.5% of patients in the surgical arm, compared with 1.1% in the percutaneous arm). Only 172 of the originally planned 328 patients were randomized; therefore, these findings are hypothesis-generating only and require replication. With regards to loss-to-follow-up, this was overall low and less common after TAVR than SAVR. A principal cause of patients being lost to follow-up is patient death. A vital-status sweep performed in the PARTNER 3 identified more patients that had died in the surgical arm. It is not known how vital-status sweeps to reduce loss-to-follow-up across all trials would have impacted our results.

Our survival analyses are not derived from true individual participant data and are dependent on the quality of the figures and extraction of the data points from these figures. However, it is notable that the fidelity of our time-to-event data reconstruction was high. Interestingly, in the unadjusted Cox models, the assumption of nonproportional hazards appeared to be violated. However, before embarking upon this analysis, we anticipated heterogeneity caused by the differential baseline hazards in the distinct randomized populations and prespecified the inclusion of a γ frailty term within the Cox models to account for heterogeneity between trials. This had the benefit of rendering the global Schoenfeld P value nonsignificant, ensuring that the Cox regression approach for survival analyses was appropriate. This highlights that the apparent nonproportional hazards identified in the unadjusted models are likely explained by the KM plots comprising multiple trials with different effect sizes and durations with individual baseline hazard functions, rather than true nonproportional hazards. Directionally consistent and statistically significant results with RMST and proportional odds regression models, 2 statistical approaches that are robust to nonproportional

CENTRAL ILLUSTRATION Transcatheter vs Surgical Aortic Valve Replacement for Aortic Stenosis in Lower-Risk Patients



Reddy RK, et al. JACC. 2025;85(9):926-940.

(Left) Summary forest plot for death, death or disabling stroke, and stroke. (Right) Kaplan-Meier plot for all-cause death at 5 years comparing transcatheter aortic valve replacement (TAVR) with surgical aortic valve replacement (SAVR). IPD = individual participant data.

hazards, add further assurances that our results and conclusions are valid.

Very long-term follow-up was absent from most trials with our analyses extended to 5-year follow-up. Three of the included trials only had 1-year follow-up. Although our analytic models take into account differential follow-up durations, longer-term follow-up is needed and is particularly important for lower-risk patients. Importantly, our results do not apply to patients with bicuspid aortic stenosis who were excluded from all trials aside from NOTION-2. In NOTION-2, the outcomes with TAVR appeared worse in the bicuspid cohort, although only 100 patients were included in this comparison. Dedicated RCTs of TAVR for bicuspid aortic stenosis are currently being planned. Our analysis could not speak to the lifetime management issues of valve durability, coronary revascularization procedures, and further aortic valve interventions.

A recently presented trial compared balloon-expandable TAVR with SAVR in 420 female patients with severe aortic stenosis.[29] Because this trial has not yet been published, we have not included these preliminary data in this analysis. Finally, the majority of patients enrolled in these trials were in their 70s, and the comparative longer-term performance of TAVR vs SAVR in younger patients remains unknown.

CONCLUSIONS

Across all available randomized data in lower-risk patients, TAVR was associated with a 20% reduction in death and 19% reduction in death or disabling stroke compared with SAVR. However, most patients included in this analysis have not yet undergone 5-year follow-up and these findings may change as further longer-term data become available. Until then, these data are informative for lower-risk patients and their treating clinicians, although longer-term follow-up is required to establish whether these trends continue and to study valve durability and other lifetime management issues, which were beyond the scope of this analysis.

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REFERENCES

1. Sá MP, Jacquemyn X, Van den Eynde J, et al. Midterm survival of low-risk patients treated with transcatheter versus surgical aortic valve replacement: meta-analysis of reconstructed time-to-event data. *J Am Heart Assoc*. 2023;12:e030012.

2. Di Pietro G, Improta R, De Filippo O, et al. Transcatheter aortic valve replacement in low surgical risk patients: an updated metanalysis of extended follow-up randomized controlled trials. *Am J Cardiol*. 2024;224:56–64.

3. Forrest JK, Deeb GM, Yakubov SJ, et al. 4-year outcomes of patients with aortic stenosis in the Evolut low risk trial. *J Am Coll Cardiol*. 2023;82(22):2163–2165.

4. Mack MJ, Leon MB, Thourani VH, et al. Transcatheter aortic-valve replacement in low-risk patients at five years. *N Engl J Med*. 2023;389(21):1949–1960. https://doi.org/10.1056/NEJMoa2307447

5. Thyregod HGH, Jørgensen TH, Ihlemann N, et al. Transcatheter or surgical aortic valve implantation: 10-year outcomes of the NOTION trial. *Eur Heart J*. 2024;45(13):1116–1124.

6. Blankenberg S, Seiffert M, Vonthein R, et al. Transcatheter or surgical treatment of aortic-valve stenosis. *N Engl J Med*. 2024;390:1572–1583.

7. Jørgensen TH, Thyregod HGH, Savontaus M, et al. Transcatheter aortic valve implantation in low-risk tricuspid or bicuspid aortic stenosis: The NOTION-2 Trial. *Eur Heart J*. 2024;45(37):3804–3814.

8. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*. 2021;372:n71.

9. Ahmad Y, Howard JP, Arnold AD, et al. Transcatheter versus surgical aortic valve replacement in lower-risk and higher-risk patients: a meta-analysis of randomized trials. *Eur Heart J*. 2023;44:836–852.

10. Sterne JAC, Savović J, Page MJ, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *BMJ*. 2019;366:l4898.

11. Guyatt GH, Oxman AD, Vist GE, et al. GRADE: an emerging consensus on rating quality of

evidence and strength of recommendations. *BMJ*. 2008;336:924–926.

12. Langan D, Higgins JPT, Jackson D, et al. A comparison of heterogeneity variance estimators in simulated random-effects meta-analyses. *Res Synth Methods*. 2019;10:83–98.

13. Higgins JPT, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med*. 2002;21:1539–1558.

14. Guyot P, Ades AE, Ouwens MJNM, Welton NJ. Enhanced secondary analysis of survival data: reconstructing the data from published Kaplan-Meier survival curves. *BMC Med Res Methodol*. 2012;12:9.

15. Wei Y, Royston P. Reconstructing time-to-event data from published Kaplan-Meier curves. *Stata J*. 2017;17:786–802.

16. Balan TA, Putter H. A tutorial on frailty models. *Stat Methods Med Res*. 2020;29:3424–3454.

17. GRAMBSCH PM, THERNEAU TM. Proportional hazards tests and diagnostics based on weighted residuals. *Biometrika*. 1994;81:515–526.

18. Royston P, Parmar MKB. Restricted mean survival time: an alternative to the hazard ratio for the design and analysis of randomized trials with a time-to-event outcome. *BMC Med Res Methodol*. 2013;13:152.

19. Page MJ, Higgins JP, Sterne JA. Assessing risk of bias due to missing results in a synthesis. In: *Cochrane Handbook for Systematic Reviews of Interventions*. John Wiley and Sons, Ltd; 2019:349–374.

20. Toff WD, Hildick-Smith D, Kovac J, et al, UK TAVI Trial Investigators. Effect of transcatheter aortic valve implantation vs surgical aortic valve replacement on all-cause mortality in patients with aortic stenosis: a randomized clinical trial. *JAMA*. 2022;327:1875–1887.

21. Guerrero Mayra. TAVR in young patients: current treatment patterns in the US. TCTMD. Accessed January 18, 2025. https://www.tctmd.com/slide/tavr-young-patients-current-treatment-patterns-us

22. Sá MP, Jacquemyn X, Sun T, et al. Late outcomes of permanent pacemaker implantation after TAVR: meta-analysis of reconstructed time-to-event data. *J Soc Cardiovasc Angiogr Interv*. 2022;1:100434.

23. Sá MP, Jacquemyn X, Van den Eynde J, et al. Impact of paravalvular leak on outcomes after transcatheter aortic valve implantation: meta-analysis of Kaplan-Meier-derived individual patient data. *Struct Heart J Heart Team*. 2023;7:100118.

24. Sá MP, Jacquemyn X, Van den Eynde J, et al. Impact of prosthesis-patient mismatch after surgical aortic valve replacement: systematic review and meta-analysis of reconstructed time-to-event data of 122 989 patients with 592 952 patient-years. *J Am Heart Assoc*. 2024;13:e033176.

25. Sá MP, Jacquemyn X, Van den Eynde J, et al. Impact of prosthesis-patient mismatch after transcatheter aortic valve replacement: meta-analysis of kaplan-meier-derived individual patient data. *JACC Cardiovasc Imaging*. 2023;16:298–310.

26. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). Chapter 10: Analysing data and undertaking meta-analyses Accessed August 14, 2024. https://training.cochrane.org/handbook/current/chapter-10

27. Barili F, Brophy JM, Ronco D, et al. Risk of bias in randomized clinical trials comparing transcatheter and surgical aortic valve replacement: a systematic review and meta-analysis. *JAMA Netw Open*. 2023;6:e2249321.

28. Kedhi E, Hermanides RS, Dambrink JE, et al. TransCatheter aortic valve implantation and fractional flow reserve-guided percutaneous coronary intervention versus conventional surgical aortic valve replacement and coronary bypass grafting for treatment of patients with aortic valve stenosis and complex or multivessel coronary disease (TCW): an international, multicentre, prospective, open-label, non-inferiority, randomised controlled trial. *Lancet*. 2025;404(10471):2593–2602.

29. Eltchaninoff H. Transcatheter versus surgical aortic valve replacement in women with severe aortic stenosis. *Paper presented at*. London, England: ESC Congress; 2024.

KEY WORDS aortic surgery, aortic valve replacement, meta-analysis, randomized controlled trials, transcatheter aortic valve implantation

APPENDIX For supplemental tables and figures, please see the online version of this paper.